A Posteriori Error Estimation

Residuals, local indicators, and reliability

From a rate to an error estimate

Chapter 7 ended with an a priori statement such as

$$\|u - u_h\|_a \leq C_I \sqrt{k_1}\, h^{s-1}\, |u|_{H^s(\Omega)}\,.$$

It predicts a rate, but it does not answer two questions after a computation:

1. Is the error on this mesh small enough?
2. Which elements should be refined?

Chapter 8 replaces the unknown derivative of u by defects computed from u_h:

$$\boxed{\text{solve} \longrightarrow \text{estimate} \longrightarrow \text{mark} \longrightarrow \text{refine.}}$$

Today: derive the residual, localize it to elements and facets, and build the estimator. Marking and refinement follow in the next class.

The residual is what the discrete solution leaves unsatisfied

For the diffusion problem, let

$$a(w, v) = \int_\Omega k \nabla w \cdot \nabla v \, \mathrm{d}x, \qquad \ell(v) = \int_\Omega f v \, \mathrm{d}x - \int_{\Gamma_N} q v \, \mathrm{d}s.$$

The exact and discrete problems are

$$a(u, v) = \ell(v) \quad \forall v \in \mathcal{V}_0, \qquad a(u_h, v_h) = \ell(v_h) \quad \forall v_h \in \mathcal{V}_{h,0}.$$

Define the residual functional

$$\boxed{R(u_h; v) = \ell(v) - a(u_h, v), \qquad v \in \mathcal{V}_0.}$$

With $e = u - u_h$,

$$\boxed{R(u_h; v) = a(e, v).}$$

Therefore $R(u_h; v_h) = 0$ for every $v_h \in \mathcal{V}_{h,0}$: the residual vanishes on the test space used to compute u_h, but generally not on all of $\mathcal{V}_0$.

In the energy norm, residual size equals error size

Because $a(\cdot,\cdot)$ is an inner product,

$$|R(u_h; v)| = |a(e, v)| \leq \|e\|_a \, \|v\|_a .$$

Taking the supremum gives one inequality. Taking $v = e$ gives equality:

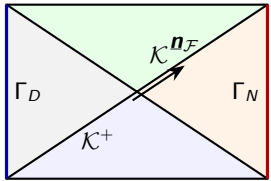
$$\boxed{\|e\|_a = \sup_{0 \neq v \in \mathcal{V}_0} \frac{|R(u_h; v)|}{\|v\|_a}.}$$

This identity is exact, but it is not yet computable:

- the maximizing direction is $v = e$, which is unknown;
- the supremum ranges over the infinite-dimensional space $\mathcal{V}_0$;
- the single number does not yet say where the error is located.

The next step rewrites $R(u_h; v)$ as element and facet contributions.

A conforming mesh has three facet classes



- $\mathcal{F}_h^{\mathrm{int}}$: shared by two elements $\mathcal{K}^+$ and $\mathcal{K}^-$;
- $\mathcal{F}_h^D$: boundary facets on Γ_D;
- $\mathcal{F}_h^N$: boundary facets on Γ_N.

Choose $\boldsymbol{n}_{\mathcal{F}}$ from $\mathcal{K}^+$ to $\mathcal{K}^-$. For a piecewise quantity w,

$$[\![w]\!]_{\mathcal{F}} = w^+ - w^-.$$

Reversing the element labels reverses both the jump and the normal, so the normal-flux jump used below is unchanged.

Elementwise integration by parts exposes the strong defect

Start from

$$R(u_h; v) = \int_\Omega f v \, \mathrm{d}x - \int_{\Gamma_N} q v \, \mathrm{d}s - \int_\Omega k \nabla u_h \cdot \nabla v \, \mathrm{d}x.$$

On one element $\mathcal{K}_e$,

$$\int_{\mathcal{K}_e} k \nabla u_h \cdot \nabla v \, \mathrm{d}x = - \int_{\mathcal{K}_e} \nabla \cdot (k \nabla u_h) v \, \mathrm{d}x + \int_{\partial \mathcal{K}_e} (k \nabla u_h \cdot \boldsymbol{n}_e) v \, \mathrm{d}s.$$

After summing over elements,

$$\begin{aligned} R(u_h; v) = & \sum_e \int_{\mathcal{K}_e} \big(f + \nabla \cdot (k \nabla u_h)\big) v \, \mathrm{d}x \\ & - \sum_e \int_{\partial \mathcal{K}_e} (k \nabla u_h \cdot \boldsymbol{n}_e) v \, \mathrm{d}s - \int_{\Gamma_N} q v \, \mathrm{d}s. \end{aligned}$$

The first line is the elementwise balance defect. The second line must be regrouped facet by facet.

Facet regrouping gives two flux defects

Define the computed flux $\boldsymbol{q}_h = -k\nabla u_h$. On an interior facet, the two element-boundary terms combine into

$$j_{\mathcal{F}} = -[\![k\nabla u_h \cdot \boldsymbol{n}_{\mathcal{F}}]\!]_{\mathcal{F}} = [\![\boldsymbol{q}_h \cdot \boldsymbol{n}_{\mathcal{F}}]\!]_{\mathcal{F}}.$$

On a Neumann facet,

$$j_{\mathcal{F}} = -\big(q + k\nabla u_h \cdot \boldsymbol{n}\big) = \boldsymbol{q}_h \cdot \boldsymbol{n} - q.$$

On Γ_D, $v = 0$, so there is no residual term. With

$$r_e = f + \nabla\cdot\left(k\nabla u_h\right) = f - \nabla\cdot\boldsymbol{q}_h,$$

the exact representation is

$$\boxed{R(u_h; v) = \sum_e \int_{\mathcal{K}_e} r_e v \,\mathrm{d}x + \sum_{\mathcal{F}\in\mathcal{F}_h^{\mathrm{int}}\cup\mathcal{F}_h^N} \int_{\mathcal{F}} j_{\mathcal{F}} v \,\mathrm{d}s.}$$

What the residuals measure

Element residual
$r_e = f - \nabla \cdot \boldsymbol{q}_h$
balance defect inside an element

Interior facet
$j_{\mathcal{F}} = [\![\boldsymbol{q}_h \cdot \boldsymbol{n}_{\mathcal{F}}]\!]$
flux leaving one element does not enter the next

Neumann facet
$j_{\mathcal{F}} = \boldsymbol{q}_h \cdot \boldsymbol{n} - q$
computed flux misses prescribed flux

For $p = 1$ and elementwise-constant k, ∇u_h is constant on each element, so $\nabla \cdot (k \nabla u_h) = 0$ and $r_e = f$. The computed gradient information then appears in the facet jumps.

Galerkin orthogonality balances the two kinds of defects on every free basis function ϕ_i:

$$\sum_e \int_{\mathcal{K}_e} r_e \phi_i \, \mathrm{d}x + \sum_{\mathcal{F}} \int_{\mathcal{F}} j_{\mathcal{F}} \phi_i \, \mathrm{d}s = 0.$$

Neither residual must vanish separately.

Activity: residuals for the uniformly loaded bar

Let $L = AE = f = 1$, $\overline{P} = 0$, and use a uniform mesh of size h with linear elements. The exact solution is

$$u(x) = x - \frac{x^2}{2},$$

and Chapter 7 showed $u_h = \mathcal{I}_h u$.

Work with a neighbor. For element $\mathcal{K}_e = (x_e, x_{e+1})$:

1. Find $u_h'|_{\mathcal{K}_e}$ using the chord slope of u.
2. Compute

$$r_e = f + (AEu_h')', \quad j_i = (AEu_h')(x_i^+) - (AEu_h')(x_i^-), \quad j_{N+1} = \overline{P} - (AEu_h')(L^-).$$

3. For an interior hat function ϕ_i, verify that its element and facet residual contributions cancel.

The nodal values are exact. Predict: do all residuals vanish?

Activity debrief: exact nodes do not imply zero residual

The chord slope equals the exact derivative at the midpoint $\bar{x}_e$:

$$u_h'|_{\mathcal{K}_e} = u'(\bar{x}_e) = 1 - \bar{x}_e.$$

Therefore

$$\boxed{r_e = 1, \qquad j_i = -h, \qquad j_{N+1} = -\frac{h}{2}.}$$

Every residual is nonzero, although $u_h(x_i) = u(x_i)$ at every node.

For an interior hat function,

$$\int_0^1 r\phi_i \, \mathrm{d}x = h, \qquad j_i\phi_i(x_i) = -h.$$

At the last node, the two terms are $h/2$ and $-h/2$. This is residual orthogonality in local form: the discrete equations balance element and facet defects, but the defects still detect the error between nodes.

Galerkin orthogonality supplies the powers of h

The residual representation alone contains $\|e\|_{L^2(\mathcal{K}_e)}$, which gives no local mesh-size weight. For any $v_h \in \mathcal{V}_{h,0}$,

$$R(u_h; v) = R(u_h; v - v_h).$$

Choose a local approximation $v_h = \mathcal{Q}_h v$. Hypothesis (Q) requires

$$\|v - \mathcal{Q}_h v\|_{L^2(\mathcal{K}_e)} \le C_Q \, h_e \, |v|_{H^1(\omega_e)} ,$$
$$\|v - \mathcal{Q}_h v\|_{L^2(\mathcal{F})} \le C_Q \, h_{\mathcal{F}}^{1/2} \, |v|_{H^1(\omega_{\mathcal{F}})} .$$

Here ω_e and $\omega_{\mathcal{F}}$ are fixed-size patches. The operator maps $\mathcal{V}_0$ into $\mathcal{V}_{h,0}$ and uses local averages, so it is defined for H^1 functions even when nodal interpolation is not.

The factors h_e and $h_{\mathcal{F}}^{1/2}$ become the estimator weights.

Local indicators assign a computable number to each element

For each element $\mathcal{K}_e$, define

$$\eta_e^2 = h_e^2 \, \|r_e\|^2_{L^2(\mathcal{K}_e)} + \frac{1}{2} \sum_{\substack{\mathcal{F} \subset \partial \mathcal{K}_e \\ \mathcal{F} \in \mathcal{F}_h^{\mathrm{int}}}} h_{\mathcal{F}} \, \|j_{\mathcal{F}}\|^2_{L^2(\mathcal{F})} + \sum_{\substack{\mathcal{F} \subset \partial \mathcal{K}_e \\ \mathcal{F} \in \mathcal{F}_h^{N}}} h_{\mathcal{F}} \, \|j_{\mathcal{F}}\|^2_{L^2(\mathcal{F})} \, .$$

The global estimator is

$$\eta = \left(\sum_e \eta_e^2 \right)^{1/2} .$$

The factor $1/2$ assigns half of an interior facet to each adjacent element. After summing over elements, every facet is counted exactly once. A Neumann facet belongs to only one element, so it receives no factor $1/2$.

In one dimension, reliability has an explicit constant

For the bar with variable $AE(x) \geq (AE)_0 > 0$, nodal interpolation is defined for every H^1 error. The nodal terms vanish for $w = e - \mathcal{I}_h e$, leaving only element residuals. The interpolation bound and Cauchy–Schwarz give

$$\boxed{\|u - u_h\|_a \leq \frac{1}{\sqrt{(AE)_0}} \left(\sum_e h_e^2 \, \|r_e\|^2_{L^2(\mathcal{K}_e)} \right)^{1/2}.}$$

For the activity, $(AE)_0 = 1$, $r_e = 1$, $N = 1/h$, and

$$\eta_{1\mathrm{D}}^2 = Nh^3 = h^2, \qquad \eta_{1\mathrm{D}} = h.$$

Chapter 7 gave $\|e\|_a = h/\sqrt{12}$, so

$$I_{\mathrm{eff}} = \frac{\eta_{1\mathrm{D}}}{\|e\|_a} = \sqrt{12} \approx 3.46.$$

The estimator decreases at the correct rate but overestimates by a fixed factor on this mesh sequence.

Multidimensional reliability: statement and proof route

Assume Hypothesis (Q), shape regularity, and $k \geq k_0 > 0$. Then

$$\boxed{\|u - u_h\|_a \leq C_{\mathrm{rel}}\eta, \qquad C_{\mathrm{rel}} = C_Q\sqrt{\frac{N_{\mathrm{ov}}}{k_0}}.}$$

The proof uses four steps:

1. $\|e\|_a^2 = R(u_h; e) = R(u_h; e - \mathcal{Q}_h e)$;
2. insert the element-and-facet residual representation;
3. apply Hypothesis (Q), then Cauchy–Schwarz to the combined sums;
4. use finite patch overlap and $|e|_{H^1(\Omega)} \leq \|e\|_a / \sqrt{k_0}$.

The constants are independent of h, but C_Q and N_{ov} are not normally computed. Thus η controls the error up to a fixed factor; its displayed value is not automatically a certified numerical upper bound.

Definition, reliability, efficiency, and effectivity differ

Term	Claim
Estimator	η and η_e are computable definitions; a definition alone does not compare them with the error.
Reliability	$\|e\|_a \leq C_{\mathrm{rel}}\eta$. A small estimator implies a small error.
Local efficiency	$\eta_e \leq C_{\mathrm{eff}}(\|e\|_{a,\omega_e} + \mathrm{osc}_e)$. A large indicator should correspond to error on a nearby patch, up to data oscillation.
Effectivity	$I_{\mathrm{eff}} = \eta / \|e\|_a$, evaluated when the exact error is known, measures the numerical scale of the estimator.

Today we proved the one-dimensional reliability result and stated the multidimensional result under Hypothesis (Q). Efficiency is a desired property and a definition here; no efficiency theorem is being claimed for this candidate estimator without a matching source and precise oscillation definition.

What to retain and what comes next

- The residual is the error measured through the bilinear form: $R(u_h; v) = a(u - u_h, v)$.
- Elementwise integration by parts separates element balance defects from interior and Neumann flux defects.
- Galerkin orthogonality allows subtraction of a local approximation; this produces the powers of the local mesh size.
- η_e localizes the estimator and η combines the elements. Reliability means the estimator controls the energy error up to a mesh-independent constant.

Exit question. For linear elements with constant k and $f = 0$ on every element, can the estimator still be nonzero? Identify the term that remains.

Next class. Complete the multidimensional proof, compare marking strategies, and connect the indicators to conforming local refinement.